

Optical Isomerism — Class Notes from Pankaj Sir (PW Lakshya JEE 2027)

(Note: Based strictly on the provided PDFs, these notes cover stereocenters, chirality, and elements of symmetry (POS, COS, AOS, AAOS). Advanced topics requested in your prompt such as enantiomers, diastereomers, meso compounds, racemic mixtures, R/S configuration, and +/- rotation were not present in these specific class slides and are therefore excluded to adhere to your strict instructions.)

1. Chiral Centers & Stereocenters

Definition of Chiral Centers

- **Chiral Centers** are also referred to as **Stereogenic centers** or **Asymmetric centers** [1].
- **Condition:** A chiral center is an sp^3 hybridized tetrahedral atom attached to '**4**' **different groups** [1].
- **Formula/Shortcut:** The atom must satisfy the condition of having **4σ Bonds + 0 Lone Pairs = 4 OR 3σ Bonds + 1 Lone Pair = 4** [1].
- **Important Condition for Lone Pairs:** If a lone pair is considered, it must be **Localized** (not involved in resonance). If the lone pair is delocalized due to resonance (e.g., next to a double bond), the atom becomes sp^2 hybridized and is **not** chiral [1, 2].
- **Eligible Atoms:** Chiral centers are not limited to Carbon; they can also be Nitrogen (N), Oxygen (O), Sulfur (S), or Phosphorus (P) [1].

Tricks and Rules for Chirality

- **"Jo chiral na ho woh Achiral":** If a molecule does not meet the conditions for chirality, it is Achiral [3].
- **Isotopes act as different groups:** Carbon attached to Hydrogen (H), Deuterium (D), and Tritium (T) alongside another group is chiral because H, D, and T are considered different groups [3].
- **Positive Ions:** Ions like O^+ or N^+ can be chiral if they have 4 distinct attachments (e.g., O^+ with a Methyl, Ethyl, Hydrogen, and 1 localized lone pair makes 4 different groups, sp^3) [2].
- **Allenes ($C = C = C$):** Pankaj Sir highlighted that allenes with an **even** number of double bonds are non-planar, while those with an **odd** number are planar (sp^2 , sp , sp^2 hybridization) [4].
- **E-Isomer Priority Rule:** When identifying E isomers, priority is given based on atomic attachments, noted as the shortcut **"Real > hypo"** (e.g., a real Carbon-Carbon bond has

higher priority than a hypothetical one drawn from resonance/pi bonds) [5].

Definition of Stereo Centres (S.C.)

- **Definition:** "The atoms whereby changing two groups a new stereo isomer is formed" [6].
- **Calculation Formula:**

$$S.C. = \text{Chiral Centers} + (2 \times \text{per Geometrical Isomerism Exhibited})$$
 [6]. This applies to systems like $C = C$, $C = N$, and $N = N$ [6].
- **Key True/False Conditions:**
 - All chiral centres are stereo centres (**True**) [6].
 - All stereo centres are chiral centres (**False**) [6].

2. Elements of Symmetry

Plane of Symmetry (POS) or σ_n

- **Definition:** An imaginary plane which can divide a molecule into two equal parts, where these equal parts are mirror images of each other [7].
- **P.P. (Popu Point / Maha Important):** $POS \geq 1$ ho sakti hai (A molecule can possess one or multiple planes of symmetry) [7].
- **Alphabet Examples:** Letters like **A, B, C, D, E, H** have a Plane of Symmetry [7]. Letters like **F, G** do not [7].
- **Standard Molecular POS Values to Remember:**
 - CH_4 : 6 POS [8].
 - CH_3Cl : 3 POS [8].
 - CH_2Cl_2 : 2 POS [8].
 - $CHCl_3$: 3 POS [8].
 - CCl_4 : 6 POS [8].
 - Benzene (C_6H_6): 7 POS [9].
 - Trans-1,2-dichloroethene: 1 POS [10].

Centre of Symmetry (COS) or Centre of Inversion (C_i)

- **Definition:** An imaginary point from which, if opposite lines are drawn, the same group will meet [11].
- **Pankaj Sir's Diagram Trick:** Sir illustrated this by drawing a central point inside a rhombus and drawing opposite dotted lines extending outward; both lines met the word "Pankaj", proving the center of symmetry [11].

- **P.P. (Popu Point): COS always 1** (A molecule can only ever have a maximum of 1 Center of Symmetry) [11].
- **Key Examples:**
 - Benzene: 1 COS [12].
 - *p*-Dichlorobenzene has a COS, whereas *o*- and *m*-dichlorobenzene do not [13].
 - Fumaric acid (trans-butenedioic acid) has a COS, whereas Maleic acid (cis-butenedioic acid) does not [14, 15].
 - CH_4 , CH_3Cl , CH_2Cl_2 , $CHCl_3$, and CCl_4 all have **NO** COS [16].

Axis of Symmetry (AOS) or Line of Symmetry (C_n)

- **Definition:** The imaginary axis along which, if a molecule is rotated by $360^\circ/n$, the same picture of the molecule reappears [17].
- **Formula:** $\theta = \frac{360^\circ}{n}$ or $n = \frac{360^\circ}{\theta}$ [17].
- **Rules & Shortcuts:**
 - **C_1 is discarded:** C_1 represents a 360° rotation, which "Always hoti hai" (always exists for any object), so it is ignored [17].
 - Valid axes start at $C_2(\theta = 180^\circ)$, $C_3(\theta = 120^\circ)$, $C_4(\theta = 90^\circ)$, etc. [17].
 - **P.P. (Popu Point):** $AOS \geq 1$ ho sakti hai (Multiple axes can exist) [18].
 - **Trick for primary axis:** Always "Choose highest 'n' or Lowest ' θ '" to define the primary axis of a compound [19].
- **Examples:**
 - BCl_3 has one C_3 axis and three C_2 axes [18].
 - Benzene has a primary C_6 axis ($\theta = 60^\circ$) [19].
 - Bromobenzene has a C_2 axis [19].

Alternate Axis of Symmetry (AAOS) or Rotation-Reflection AOS (S_n)

- **Definition:** A complex element of symmetry where the object is **first rotated** around an axis by $360^\circ/n$, followed by **reflecting** the original object through a mirror plane **perpendicular** to this rotational axis [20].
- If the resulting image matches the original starting molecule, it possesses an S_n axis [20, 21].

3. Notable Examples and Exam Tips

"KBCC" Standard Values for 3D/Cage Molecules (Google Search structures)

Pankaj Sir highlighted specific highly symmetric cage molecules that are important for objective questions: * **Adamantane:** 6 POS [22-24]. * **Cubane (C_8H_8):** 9 POS [22, 24, 25] and

1 COS [26]. * **Cyclohexane (Standard Chair conformation)**: 3 POS [22, 23]. * **cis-1,3,5-trimethylcyclohexane (most stable chair)**: 3 POS [22, 23]. * **Tetrahedrane (C_4H_4)**: 6 POS [24, 25, 27]. * **Prismane (C_6H_6)**: 4 POS [24, 25, 27].

Number of Chiral Centers in Major Pharmaceutical/Complex Compounds

Sir routinely used complex drug and natural molecules to practice identifying chiral centers. * **Allegra**: 1 chiral center [28]. * **Penicillin**: 3 chiral centers [29, 30]. * **Cocaine**: 4 chiral centers [31]. * **Sativene**: 5 chiral centers [32]. * **Cortisone**: 6 chiral centers [33, 34]. * **Testosterone**: 6 chiral centers [35, 36]. * **Fluorometholone**: 8 chiral centers [37]. * **Reserpine (Antipsychotic drug)**: 8 chiral centers [38, 39]. * **Sucrose (D(+))**: 9 chiral centers [40, 41]. * **Maltose**: 9 chiral centers [42, 43].

Symbols and Visual Notation Used by Sir

- **Stars (*)**: Sir heavily utilized green and orange star symbols drawn directly onto chemical structures to denote the exact location of chiral carbon atoms [2, 3, 28, 31, 32, 37, 44-50].
- **Wedge/Dash Notation**: Extensively used to evaluate POS and COS. For example, in cyclobutane, if two Bromine atoms are on the same side (both wedge), the molecule has a POS but no COS. If they are on opposite sides (one wedge, one dash in trans position), the molecule has a COS but no POS [51, 52].

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